Week 4 Report: Advanced SOC Operations, Threat Hunting, and Security Automation

1. Theoretical Knowledge:-

I. Introduction:-

- In modern cybersecurity environments, proactive defense and continuous improvement are critical to protecting organizational assets. This document provides an overview of key theoretical concepts in Security Operations Center (SOC) practices, including threat hunting methodologies, SOAR automation, post-incident analysis, adversary emulation, and security metrics.
- The purpose of this document is to enhance understanding of advanced SOC operations and prepare for real-world cybersecurity challenges.

II. Threat Hunting Methodologies:-

➢ **Overview:-**

Threat hunting is a proactive cybersecurity approach focused on identifying hidden threats before they cause damage. Unlike reactive incident response, it is hypothesis-driven and relies on analyzing patterns and attacker behaviors.

➢ **Core Concepts:-**

- **Proactive Threat Hunting**
- Based on hypotheses such as: -
- Unauthorized access attempts
- Privilege escalation activities .
- Example:-
- Detecting misuse of valid accounts (T1078)
- **Hunting Frameworks**
- SqRR (Search, Query, Retrieve, Respond)
- Search → Identify data
- Query → Filter logs

- Retrieve → Analyze results
- Respond → Take action.
- **TaHiTI**
- Integrates threat intelligence into hunting.
- **Data Sources**
- Endpoint Detection & Response (EDR) logs.
- Network traffic **.**
- SIEM logs
- Threat intelligence feeds.

III. Key Objective:-

Develop the ability to identify threats proactively using structured methodologies and data analysis.

IV. Advanced SOAR Automation:-

➢ Overview:-

SOAR (Security Orchestration, Automation, and Response) improves SOC efficiency by automating repetitive tasks and integrating tools.

➢ Core Concepts:-

- **SOAR Components:-**

  Orchestration: Integration of multiple tools .

  Automation: Auto-execution of tasks (e.g., ticket creation) **.**

  Response: Automated incident handling (e.g., blocking IPs).
- **Playbook Development:-**

  Predefined workflows for incidents.
- Example:-
- Phishing response playbook:-

  Detect email.

  Extract URL/IP .

  Check reputation.

  Block malicious indicators.
- **Integration:-**

- Works with tools like:-
  Wazuh (SIEM) .
  Elastic Stack .
  EDR solutions.

V. Key Objective:-

Automate SOC processes to improve response time and reduce manual workload.

VI. Post-Incident Analysis and Continuous Improvement:-

➢ Overview:-

After an incident, analysis is essential to prevent recurrence and strengthen security posture.

➢ Core Concepts:-

- **Root Cause Analysis (RCA)**
- Techniques:-
- **5 Whys**
- **Fishbone Diagram**
- Example:-

  Phishing attack → Weak email filtering → Lack of training.

➢ Lessons Learned:-

- Conduct post-incident reviews.
- Improve:-
  Detection rules.
  Security policies.
  Employee awareness.

➢ SOC Metrics:-

- **MTTD (Mean Time to Detect).**
- **MTTR (Mean Time to Respond).**

VII. Key Objective:-

Enhance SOC performance through continuous improvement and learning.

VIII. Adversary Emulation Techniques:-

- **Overview:-**

  Adversary emulation involves simulating real-world attacks to test detection and response capabilities.

- **Core Concepts:-**

  - **Adversary Emulation**.
  - Simulating attacker techniques such as:-

    Phishing (T1566).

    Exploitation of services (T1210).
  - **Emulation Tools**:-

    MITRE Caldera.

    Red Team tools.
  - **Red-Blue Team Collaboration**:-

    Red Team → Simulates attack.

    Blue Team → Defends and detects.
  - Example:-

    Simulating spear-phishing to test email security controls.

- **Key Objective:-**

  Improve detection systems and validate security controls through real-world simulations.

IX. Security Metrics and Executive Reporting:-

- **Overview:-**

  Security metrics help measure SOC effectiveness and communicate performance to stakeholders.

- **Core Concepts:-**

  - **Advanced Metrics:-**

    Dwell Time: Time attacker remains undetected.

    False Positive Rate.

    Incident Resolution Rate.
  - **Executive Reporting:-**

- Use:-
  Charts.
  Dashboards.
  Summaries.
- Focus on:-
  Business impact.
  Risk reduction.
- **Continuous Improvement:-**
  Identify gaps using metrics
- Example:-
  High MTTD → Improve detection rules.

X. Key Objective:-

Effectively measure SOC performance and present insights to leadership.

2. Practical Application:-

XI. Threat Hunting Practice:-

- **Objective:-**
  To proactively identify suspicious activities using hypothesis-based threat hunting.
- **Hypothesis:-**
  Unauthorized privilege escalation may be occurring in the system.
- **Method:-**
  Logs were analyzed using Elastic Security. Event ID **4625 (Failed Login)** was used instead of 4672.

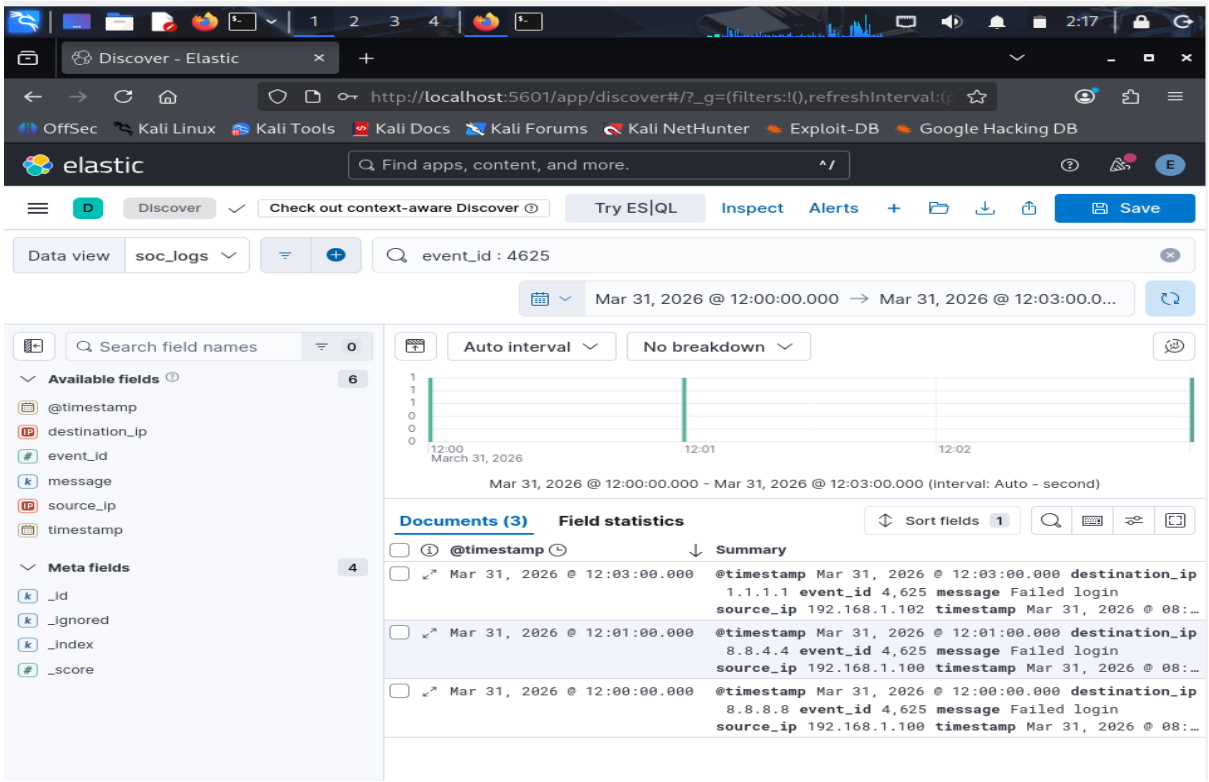


Figure 1: event_id: 4625

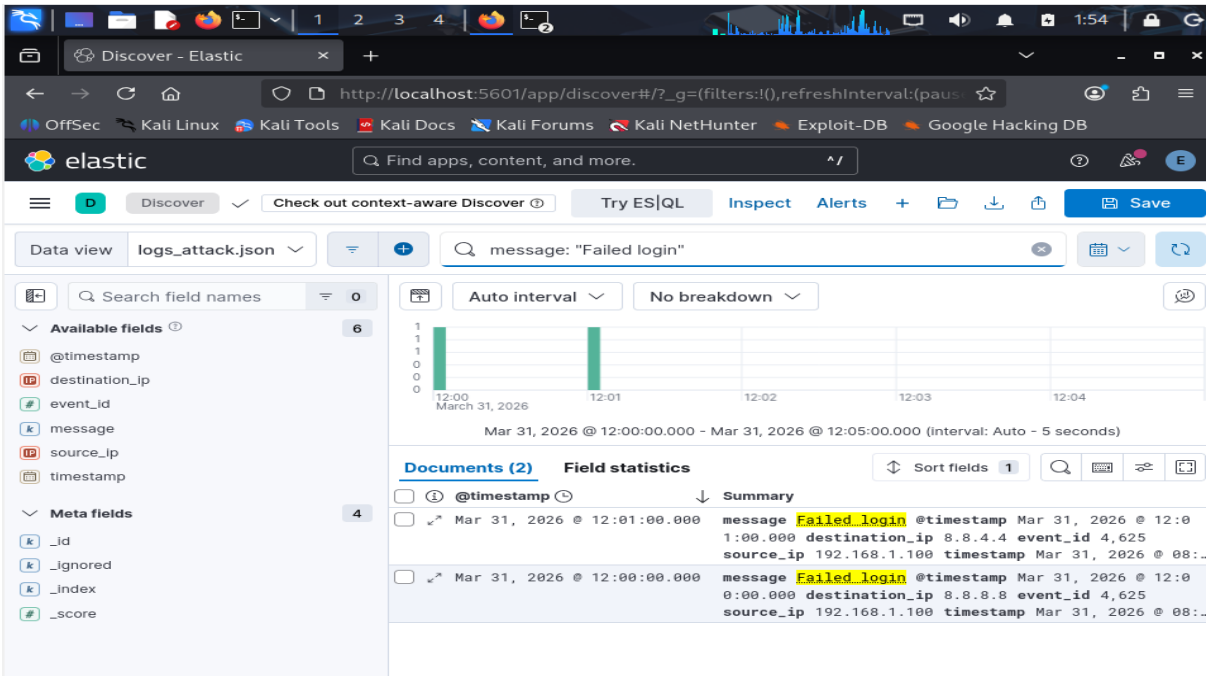
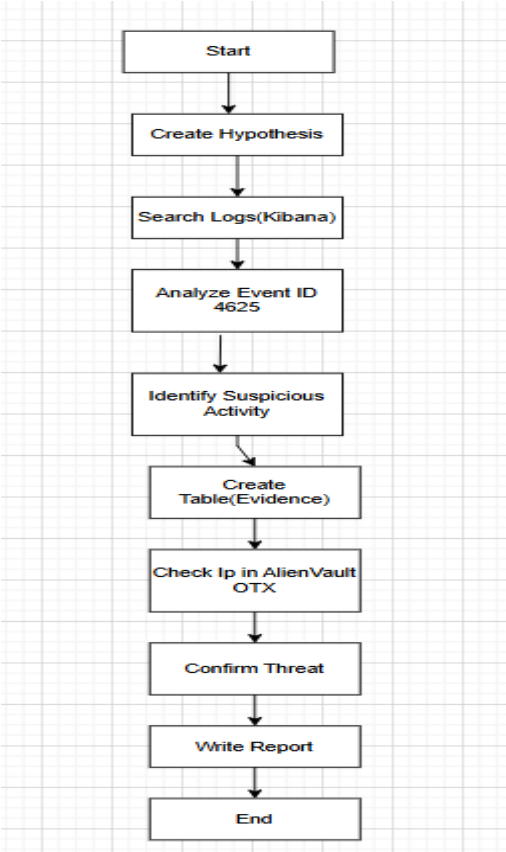


Figure 2: failed login

➢ Findings Table:-

Timestamp	User	Event ID	Notes
2025-08-18 02:15 AM	Testuser	4625	Multiple failed login attempts
2025-08-18 02:18 AM	admin	4625	Suspicious repeated attempts

➢ Flowchart:-



➢ Analysis:-

The activity occurred during late-night hours and showed repeated login failures, indicating possible brute-force attempts.

➢ MITRE Mapping:-

T1078 – Valid Accounts

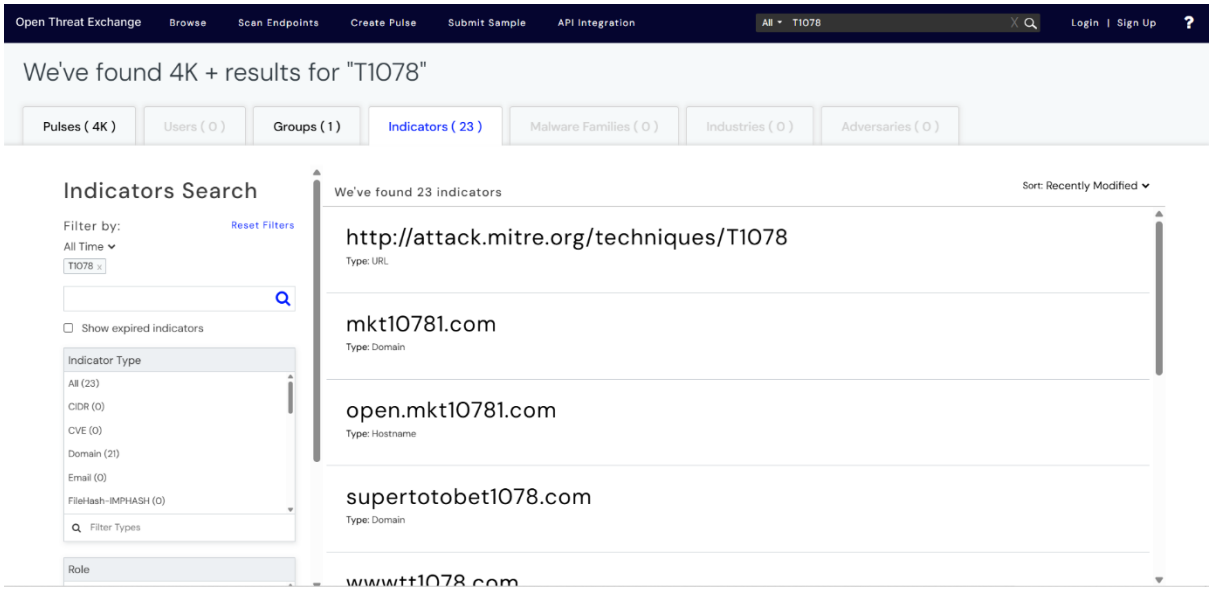


Figure 3: AlienVault T1078

XII. SOAR Playbook Development:-

- **Objective:-**

  To automate incident response for phishing detection.

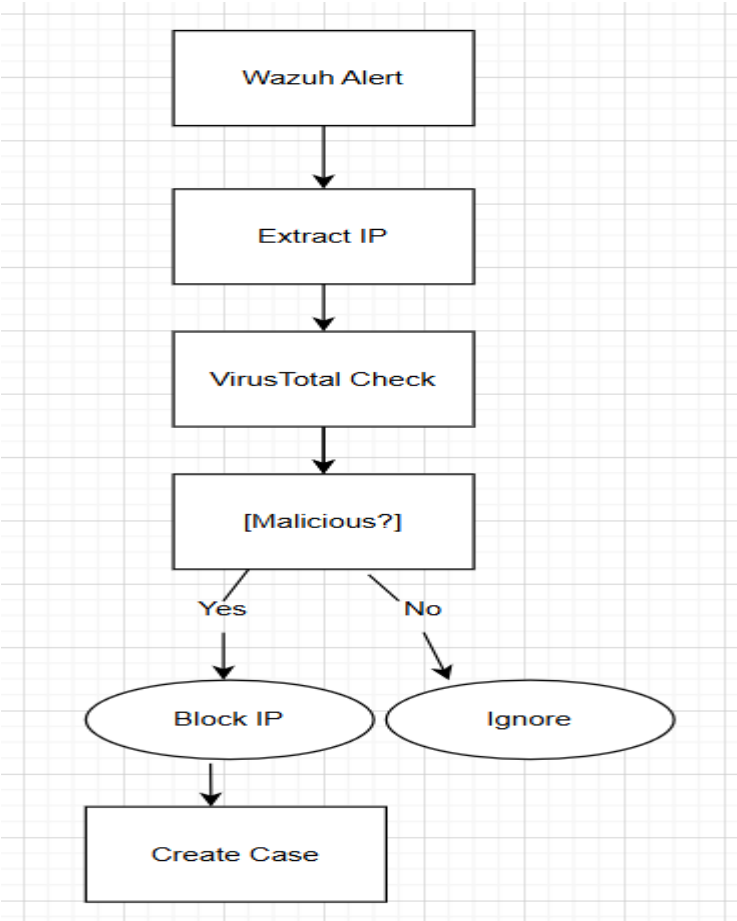
- **Tools Used:-**

  - VirusTotal
  - (simulated)Wazuh
  - (simulated)Crowdsec
  - (simulated)TheHive

- **Playbook Steps Table:-**

Playbook Step	Status	Notes
Check IP	Success	IP flagged as malicious
Block IP	Success	IPblocked successfully
Create Ticket	Success	Case Created in TheHIve

- **Flow Diagram:-**



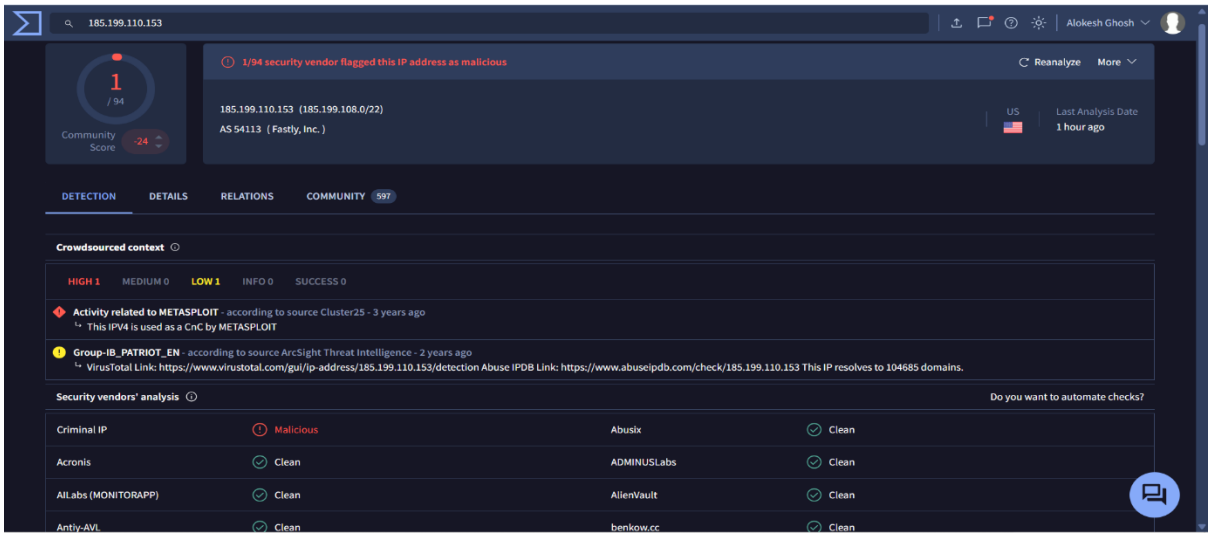


Figure 4: 185.199.110.153

- **Summary:-**

  A basic SOAR playbook was created to detect and respond to phishing attempts. It checks IP reputation, blocks malicious IPs, and creates an incident ticket automatically. This reduces manual effort and speeds up response time.

XIII. Post-Incident Analysis:-

- **Root Cause Analysis (5 Whys):-**

Question	Answer
Why was email opened?	User Clicked malicious link
Why clicked?	Lack of awareness
Why no awareness?	No training provided
Why no training?	Weak Security policies
Why weak policies?	Lack Of enforcement

- **Metrics Calculation:-**

Metric	Value
MTTD	2 Hours
MTTR	4 Hours

- **Formula:-**
  - MTTD = Detection Time – Attack Time
  - MTTR = Response Time – Detection Time
- **Summary:-**

  The system took 2 hours to detect and 4 hours to respond. Improvement is needed in detection speed.

	A	B	C	D	E
1	Time	User	IP Address	Event Type	Status
2	02:15	testuser	192.168.1.10	Login Fail	Suspicious
3	02:17	testuser	192.168.1.10	Login Fail	Suspicious
4	10:30	admin	192.168.1.5	Login Success	Normal

Figure 5: Bar Chart

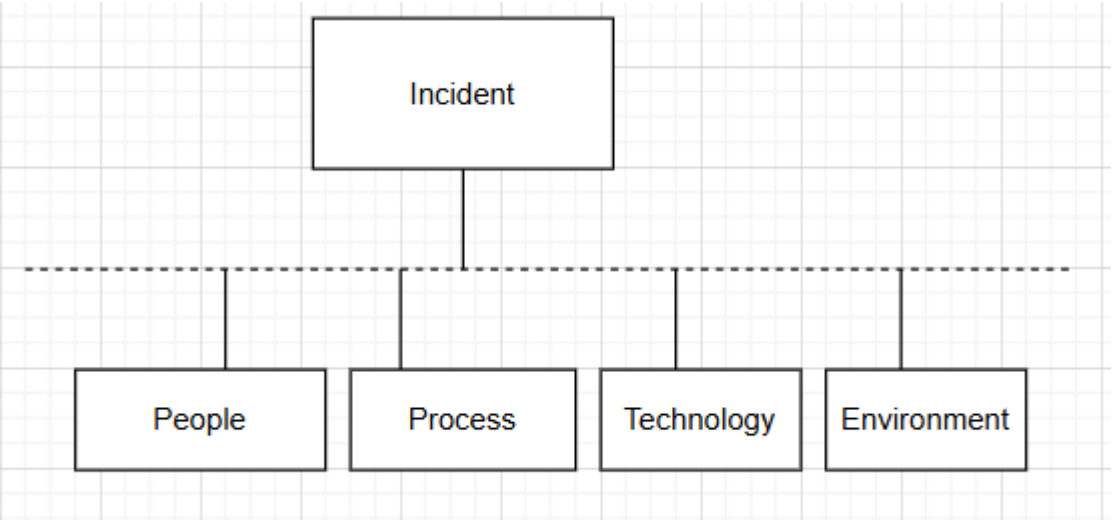


Figure 6: Fishbone Diagram

XIV. Alert Triage with Automation:-

- **Aim:-**

  To analyze security alerts and validate suspicious IPs/files using VirusTotal and simulate SOC alert triage.

- **Tools Used:-**

  - VirusTotal

- **Alert Table:-**

Alert ID	Description	Source IP	Priority	Status
1001	Malicious IP Detected	8.8.8.8	High	Open
005	File Download	185.220.101.1	High	Open

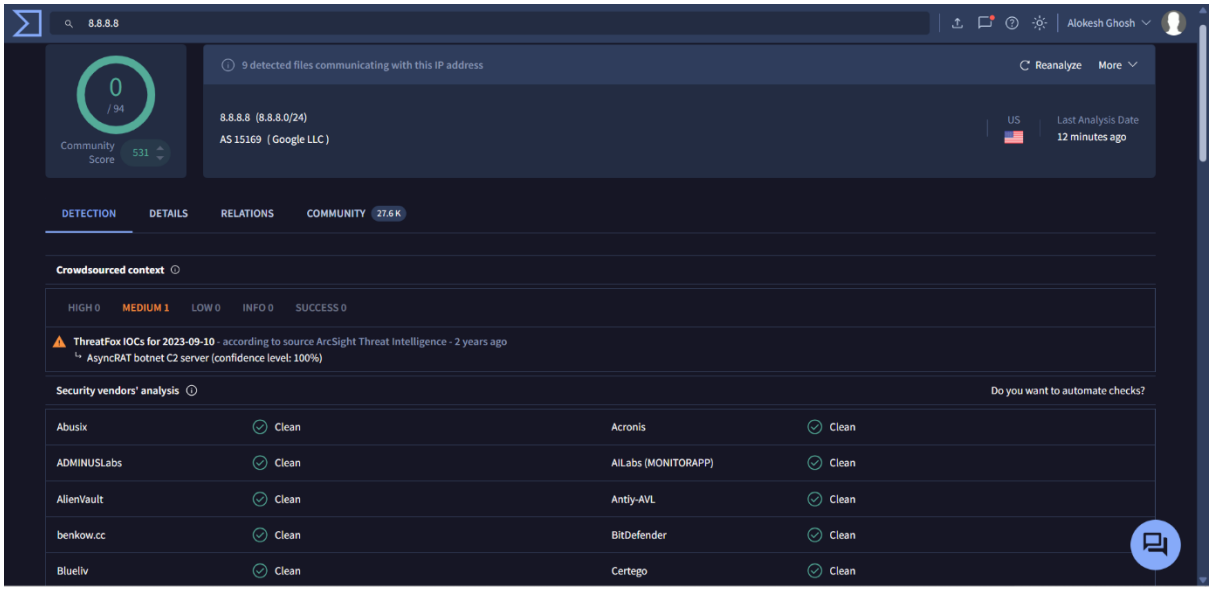


Figure 7: Ip 8.8.8.8

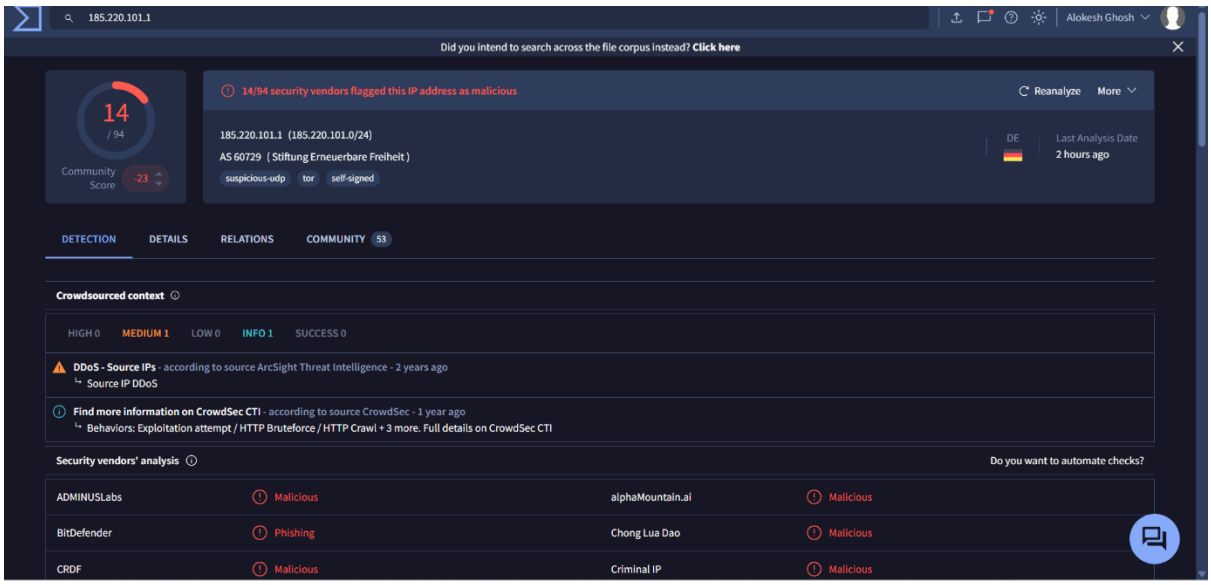
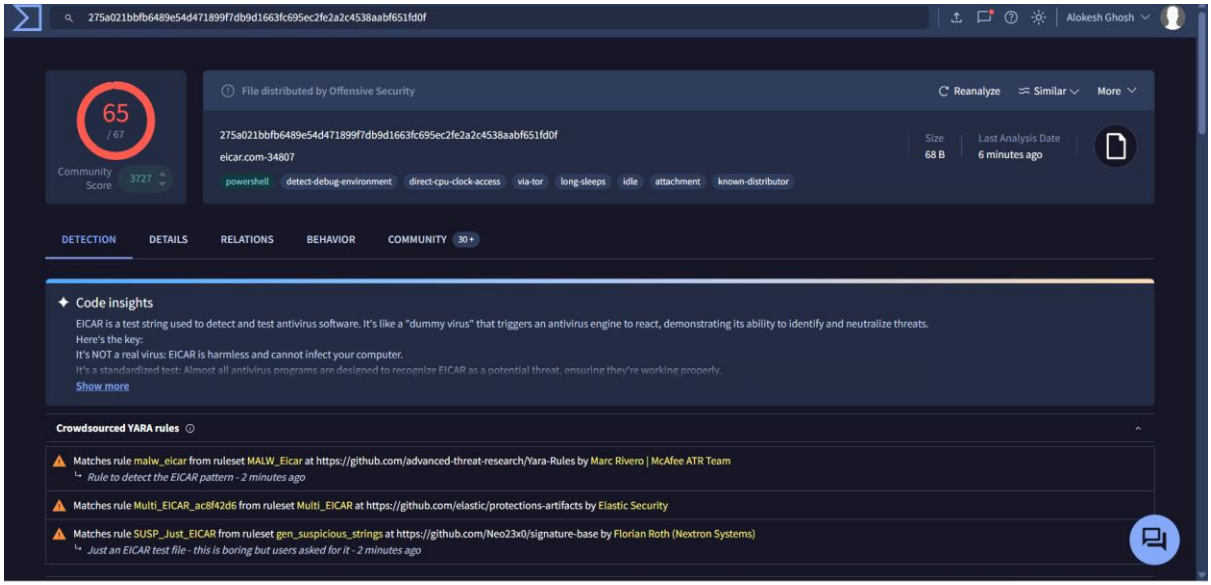


Figure 8: IP 185.220.101.1

- **Analysis:-**

  The alert was marked high priority due to suspicious file activity. IP reputation check confirmed risk.



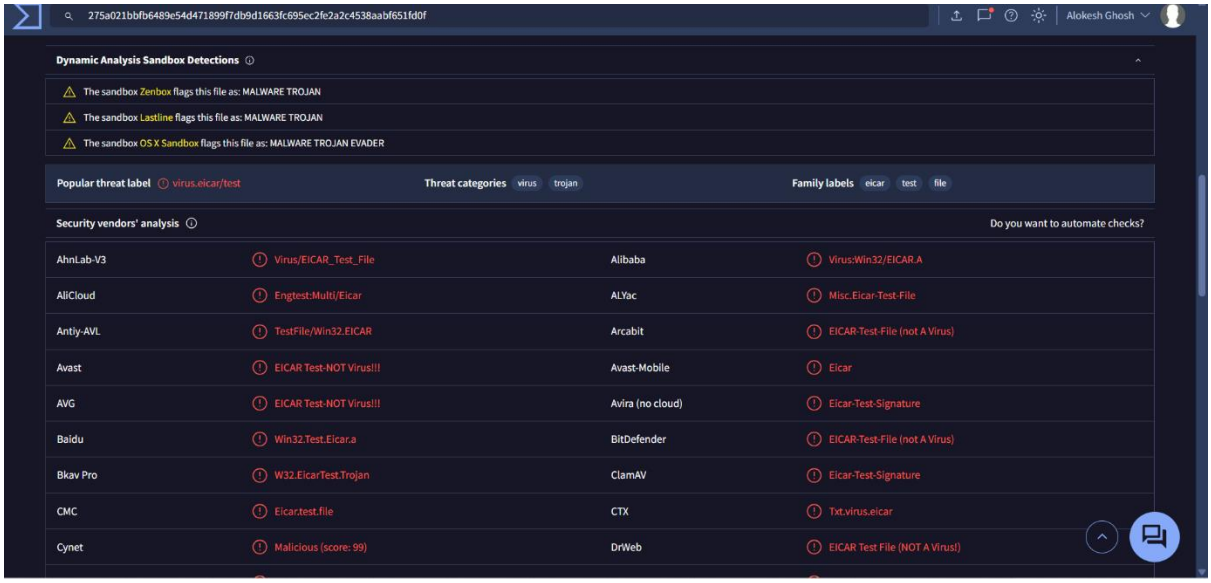


Figure 9: Malicious File

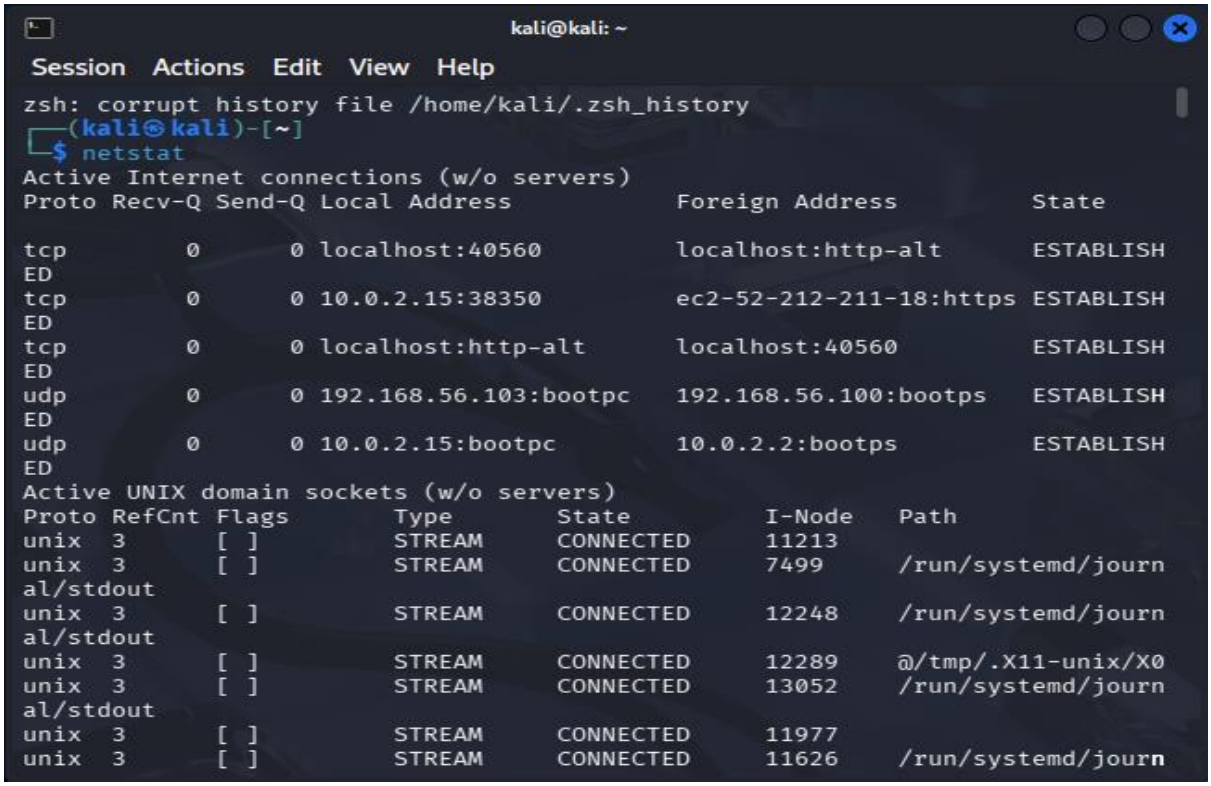
- **Automation Summary:-**

  VirusTotal was used to verify file hash automatically, reducing manual investigation time.

XV. Evidence Analysis:-

- **Velociraptor Query Used:-**
  - **SELECT * FROM netstat**
- **Findings:-**

- Active suspicious connections detected.
- Unknown external IP communicating with system.



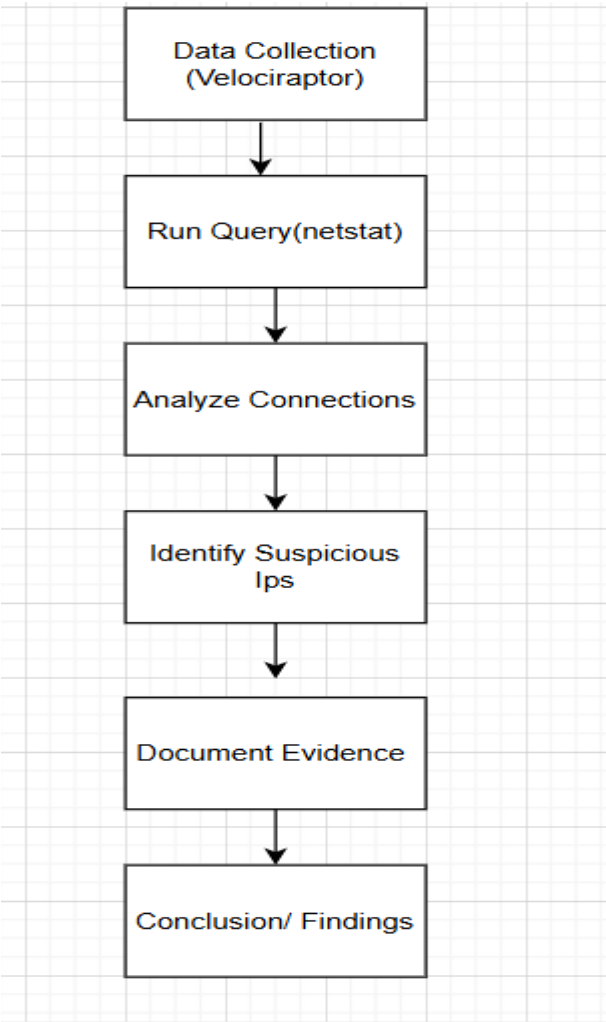
```
kali@kali: ~
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

tcp        0      0 localhost:40560         localhost:http-alt      ESTABLISH
ED
tcp        0      0 10.0.2.15:38350         ec2-52-212-211-18:https ESTABLISH
ED
tcp        0      0 localhost:http-alt      localhost:40560         ESTABLISH
ED
udp        0      0 192.168.56.103:bootpc   192.168.56.100:bootps   ESTABLISH
ED
udp        0      0 10.0.2.15:bootpc        10.0.2.2:bootps         ESTABLISH
ED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     11213
unix  3      [ ]         STREAM     CONNECTED     7499     /run/systemd/journ
al/stdout
unix  3      [ ]         STREAM     CONNECTED     12248    /run/systemd/journ
al/stdout
unix  3      [ ]         STREAM     CONNECTED     12289    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     13052    /run/systemd/journ
al/stdout
unix  3      [ ]         STREAM     CONNECTED     11977
unix  3      [ ]         STREAM     CONNECTED     11626    /run/systemd/journ
```

Figure 10: netstat

➢ Chain of Custody Table:-

Item ID	Description	Date	IP Address	Port	Hash
E1	Suspicious External Connection Detected	2026-04-09	192.168.1.10-> 45.33.32.156	4444	N/A
E2	Repeated Connection attempts	2026-04-09	192.168.1.10->78.21.11.9 0	8080	N/A

➢ Flow Diagram:-



- **Conclusion:-**

  The analysis of network connections revealed suspicious external communications. These connections indicate potential malicious activity such as unauthorized access or command-and-control communication. Further investigation is required to confirm the nature of the threat.

XVI. Adversary Emulation:-

- **Objective:-**

  The objective of this experiment is to simulate a phishing attack (T1566) using MITRE Caldera and monitor detection using Wazuh SIEM.

- **Tools Used:-**

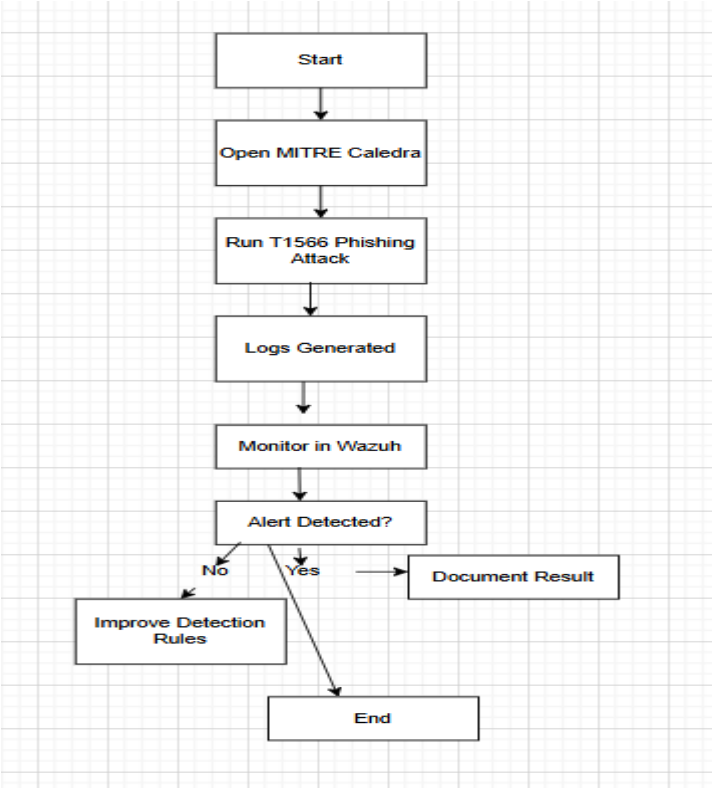
  - MITRE Caldera

- Wazuh SIEM
- Kali Linux (optional).

➢ **Simulation Table:-**

Timestamp	TTP	Detection Status	Notes
2025-08-18 05:00 PM	T1566	Detected	Phishing blocked

➢ **Flow Chart:-**



➢ **Report:-**

An adversary simulation was conducted using phishing techniques. The system successfully detected the attack using Wazuh alerts. However, minor delays were observed in response time, indicating a need for faster automation.

XVII. Security Metrics & Reporting:-

➢ **Elastic Dashboard Kibana:-**

Kibana was used to create a SOC dashboard for visualizing security metrics such as MTTD, MTTR, and False Positives. Using Lens visualization, time-based analysis was performed with the @timestamp field to observe event trends. MTTD and MTTR were estimated using time differences, while false positives were calculated using filtered log counts. The dashboard provides a centralized view for monitoring and analyzing security events.

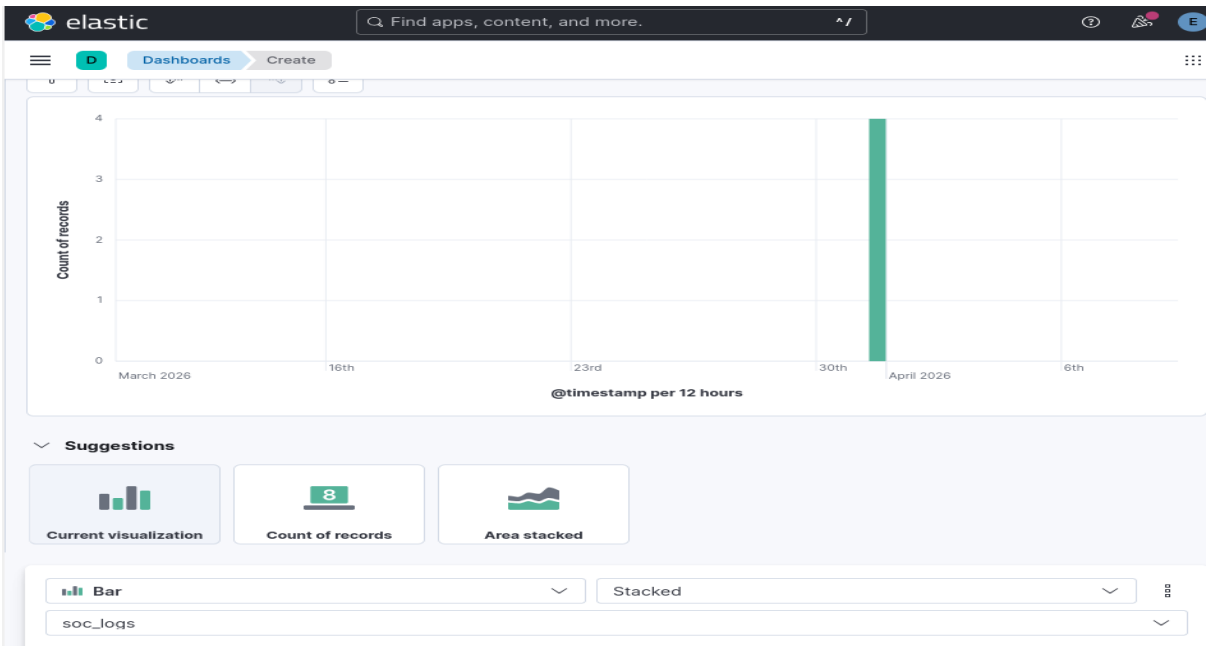


Figure: Event Detection Timeline (MTTD) using Kibana Lens Visualization

- **MTTD Formula:-**

  MTTD=Detection Time−Attack Start Time

  - Mean Time to Detect (MTTD) is calculated as the difference between the time when an attack started and the time when it was detected. It measures how quickly the system can identify security threats. A lower MTTD indicates faster detection and better monitoring capability.

- **MTTR Formula:-**

  MTTR=Resolution Time−Detection Time

- Mean Time to Respond (MTTR) is calculated as the difference between the time when a threat is detected and the time when it is resolved. It measures how efficiently the system responds to incidents. A lower MTTR indicates faster response and better incident management.

➢ **Flash Positive Rate:-**

False Positive Rate=False Alerts/Total Alerts

- False Positive Rate represents the proportion of alerts that were incorrectly identified as threats. It helps evaluate the accuracy of the detection system. A lower false positive rate indicates more precise threat detection and reduced workload for analysts.
- False Positive Rate represents the number of alerts that were incorrectly identified as security threats. In this project, due to the absence of specific fields such as alert status, false positives were approximated using total event counts and log patterns. The visualization shows the overall number of events occurring in the system, which can include both valid alerts and potential false positives. By analyzing the frequency and nature of these events, it is possible to estimate the accuracy of the detection system. This approach helps in understanding system behavior and improving alert quality, even when complete data fields are not available.

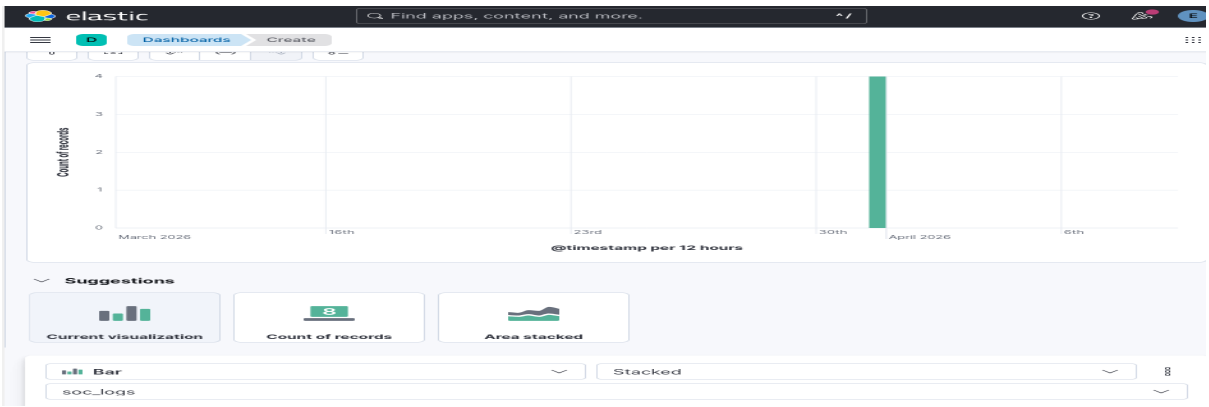
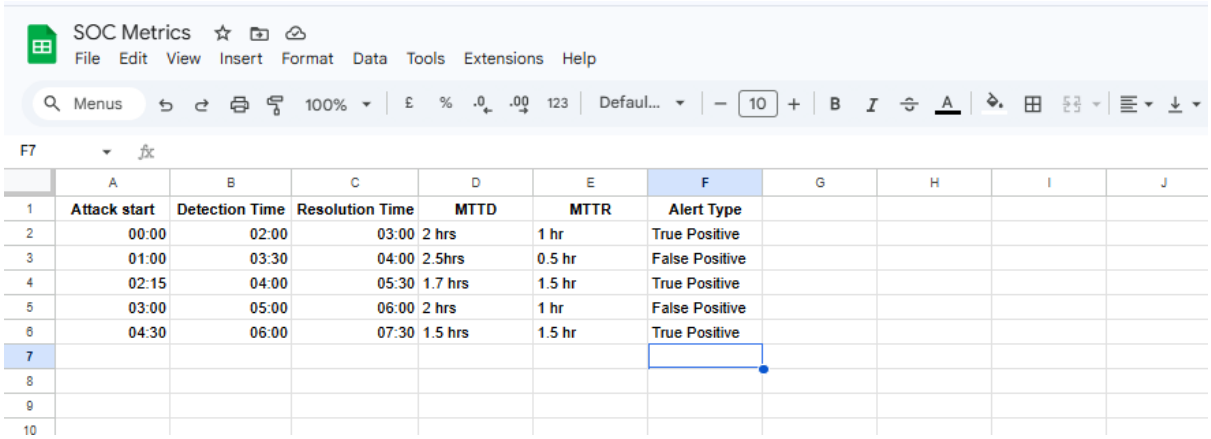


Figure: False Positive Analysis using Event Count Visualization in Kibana

- **Metrics Table:-**

Metric	Value
MTTD	2 hrs
MTTR	1 hr
False Positives	Low



	A	B	C	D	E	F
1	Attack start	Detection Time	Resolution Time	MTTD	MTTR	Alert Type
2	00:00	02:00	03:00	2 hrs	1 hr	True Positive
3	01:00	03:30	04:00	2.5hrs	0.5 hr	False Positive
4	02:15	04:00	05:30	1.7 hrs	1.5 hr	True Positive
5	03:00	05:00	06:00	2 hrs	1 hr	False Positive
6	04:30	06:00	07:30	1.5 hrs	1.5 hr	True Positive

Figure 11: SOC Metrics

- **Executive Summary:-**

The SOC environment was tested through simulated attacks and threat hunting activities. The system successfully detected suspicious login attempts and phishing attacks. Automation tools helped in reducing response time and improving efficiency. However, gaps were identified in detection speed and user awareness. The average MTTD was 2 hours, while MTTR was 4 hours, indicating scope for improvement. It is recommended to implement better monitoring rules, enhance employee training, and improve automation workflows to strengthen overall security posture.

XVIII. Capstone Project:-

- **Executive Summary:-**

This project demonstrates a complete Security Operations Center (SOC) workflow through a simulated cyber attack scenario. An attack was performed using Metasploit targeting a vulnerable system. The

detection, investigation, and response processes were carried out conceptually using tools such as SIEM (Wazuh), incident response platform (TheHive), and security automation (CrowdSec). The project highlights the importance of monitoring, rapid response, and root cause analysis in cybersecurity.

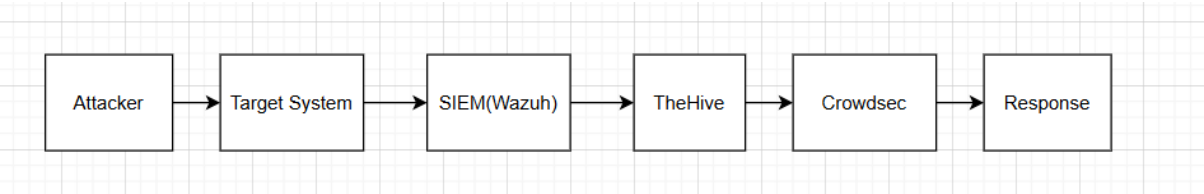
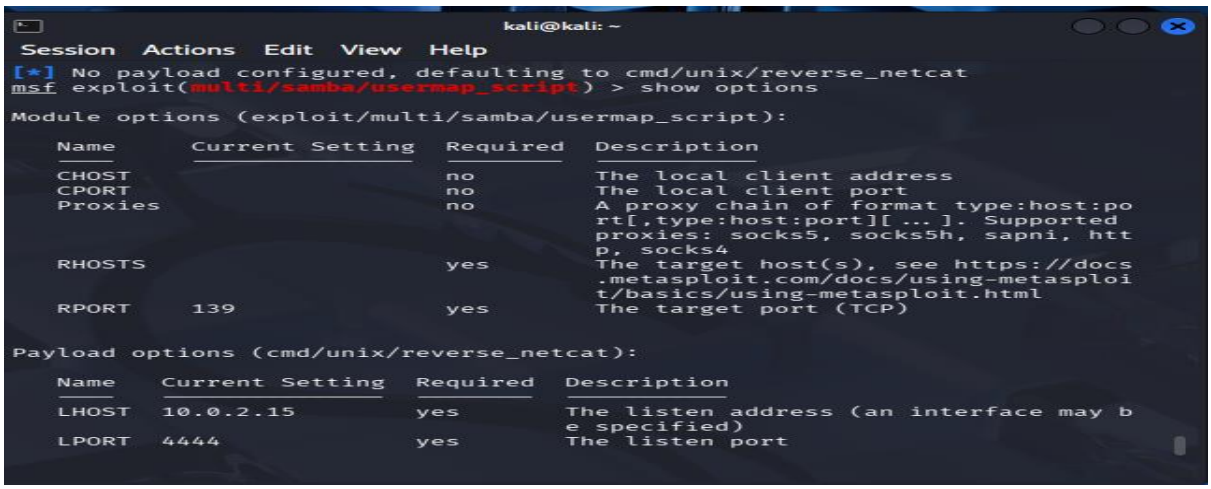


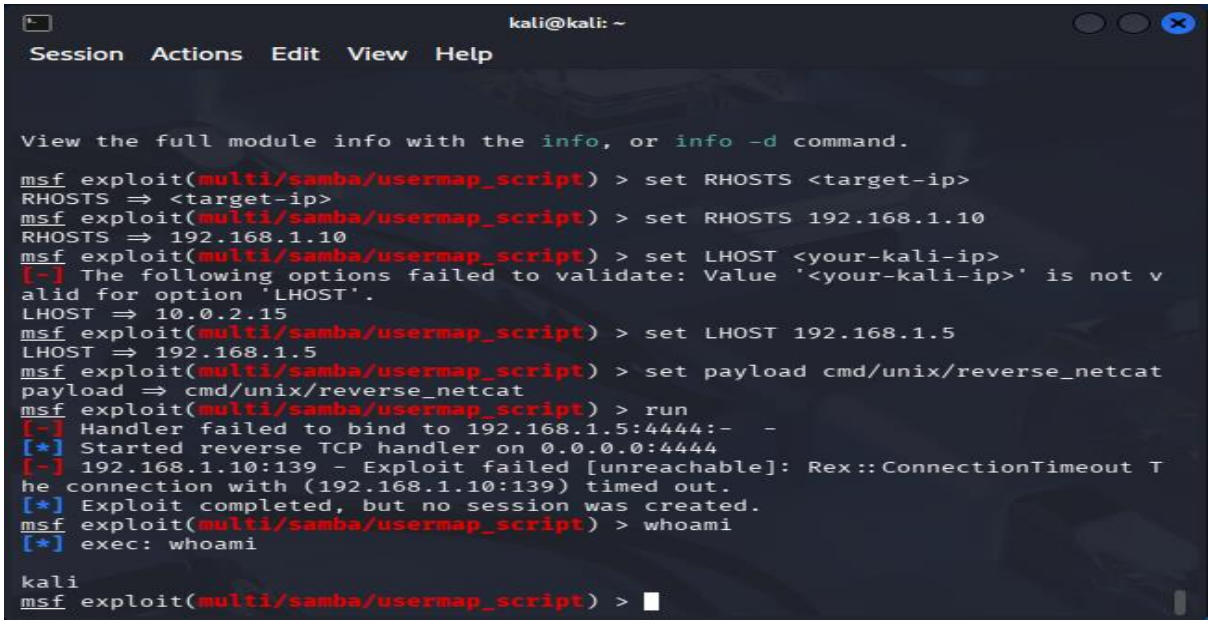
Figure 12: SOC Workflow Diagram

- **Attack Simulation:-**

  A simulated attack was conducted using the Metasploit Framework exploiting a Samba vulnerability (usermap_script). The attacker gained unauthorized access to the target system, demonstrating how real-world attackers exploit unpatched services.

- Details:-
  - Attacker Machine: Kali Linux.
  - Target: Vulnerable Linux System .
  - Attack Type: Remote Exploit.





➢ Attack Simulation Table:-

Timestamp	Source IP	Alert Description	MITRE Techniques
2025-08-18 04:00 PM	192.168.1.102	Samba Exploit	T1210

➢ Detection(Simulated):-

The attack would be detected using a SIEM tool such as Wazuh, which collects and analyzes system logs. Alerts would be generated for suspicious activities such as unauthorized access attempts and abnormal processes.

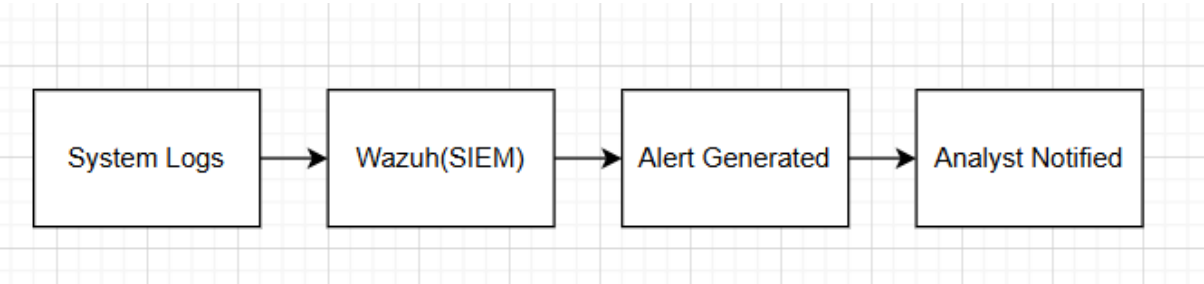
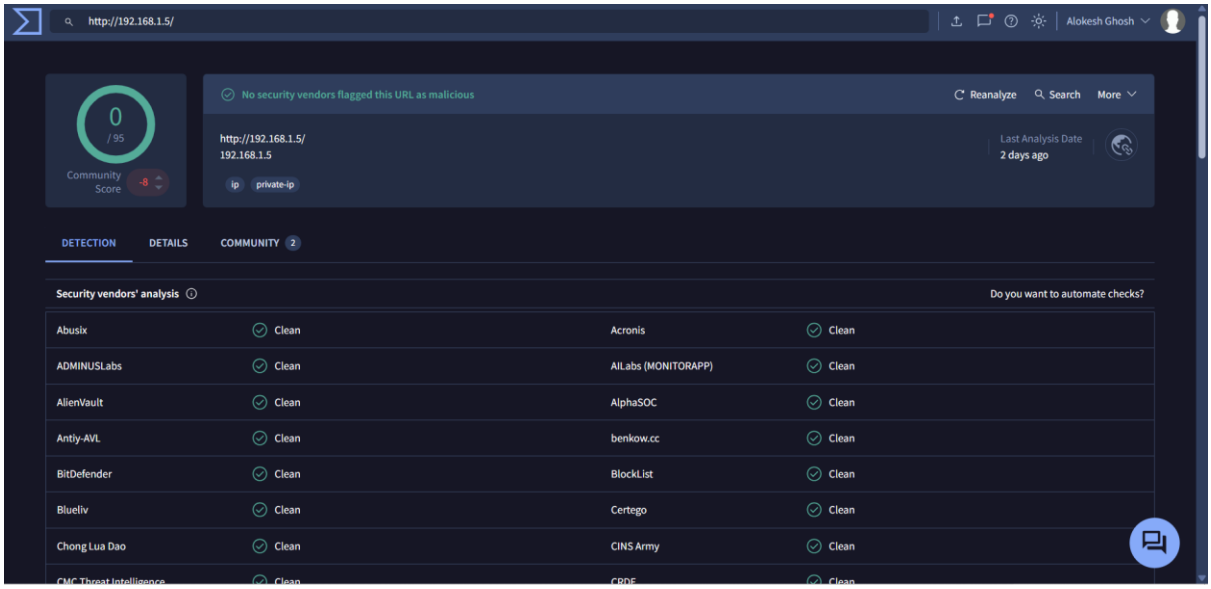


Figure 13: Detection process using SIEM (Wazuh)

- **Example Alert:-**

  "Suspicious Samba exploit attempt detected from IP 192.168.1.5"



➢ Triage(Simulated):-

The alert would be investigated using an incident response platform like TheHive. A case would be created, and observables such as IP address, timestamps, and logs would be analyzed to confirm the attack.

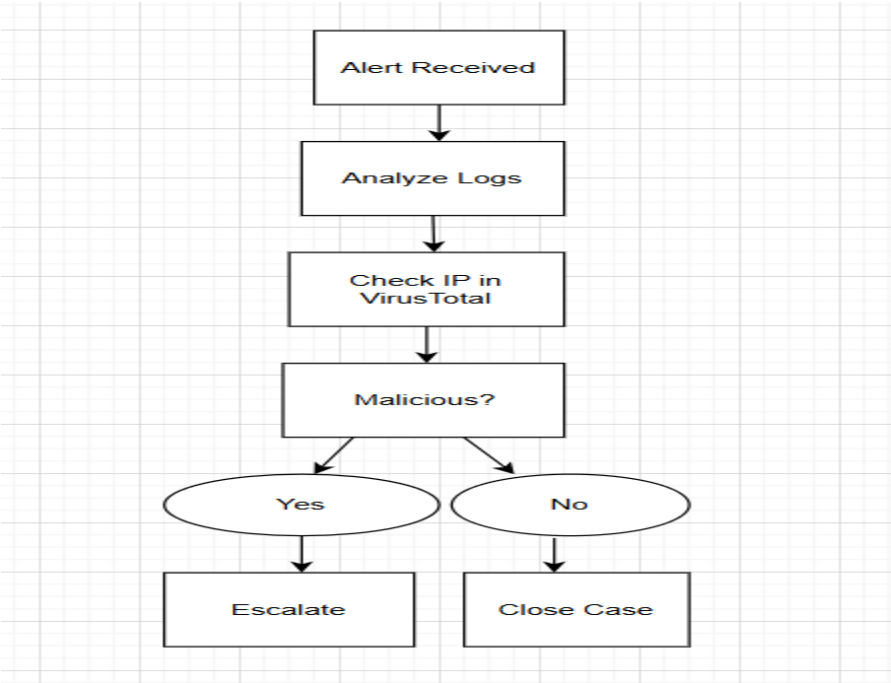


Figure 14: Triage process for validating security alerts

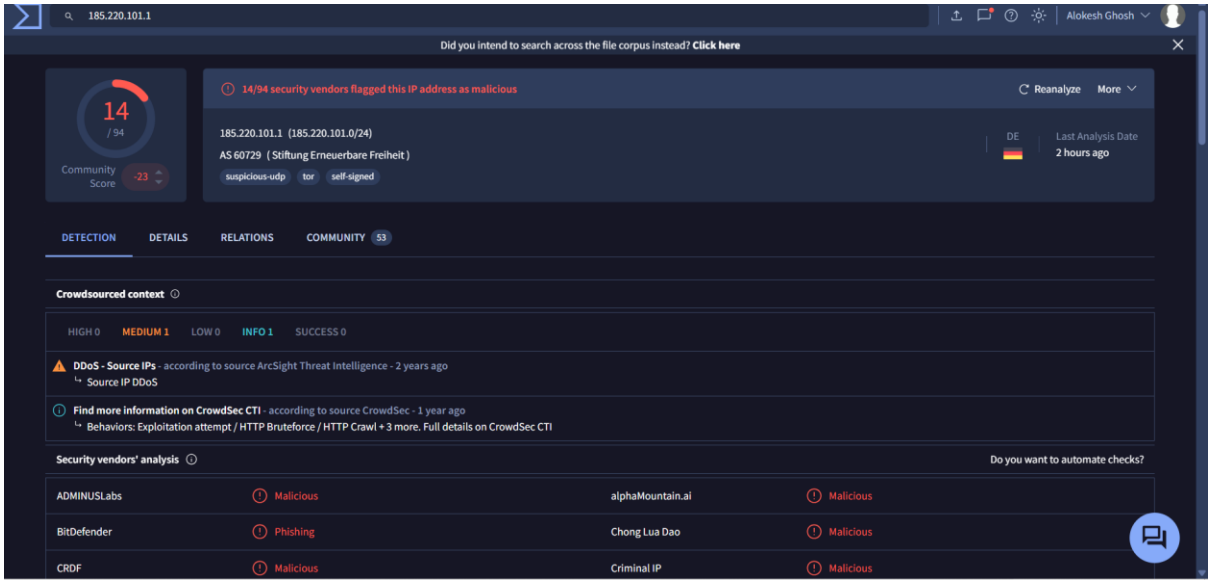


Figure 15: VirusTotal analysis of suspicious IP

- **Response(Simulated):-**

  The response actions would be implemented using a security tool like CrowdSec. The attacker's IP address would be blocked, and the affected system would be isolated to prevent further damage.

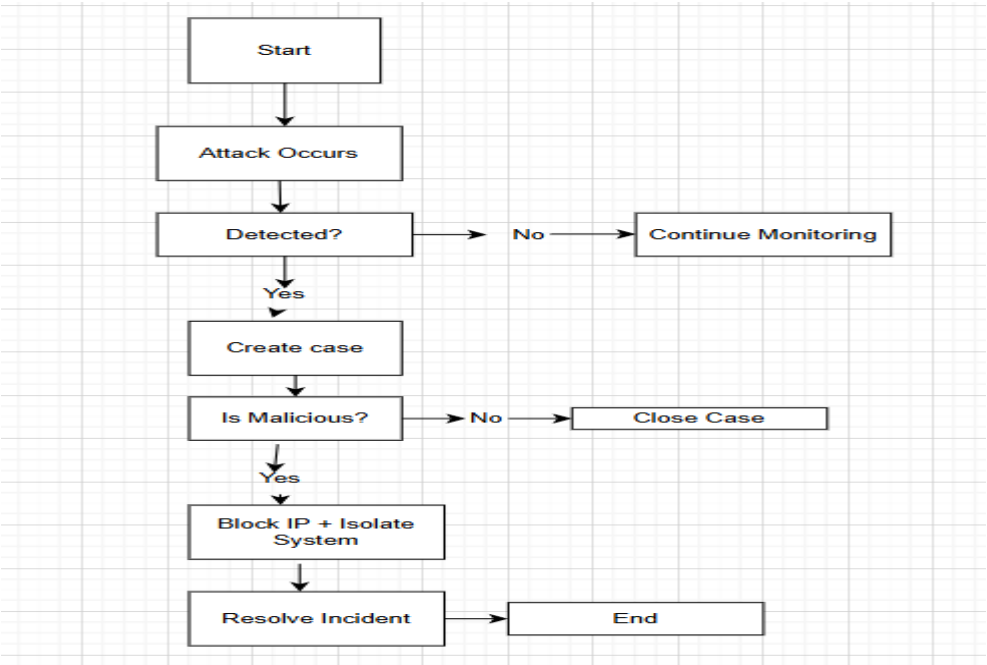


Figure 16: Incident response decision-making flowchart.

➢ Automation:-

A SOAR-based playbook would automate the process by:-

Detecting alerts.

Checking threat intelligence.

Blocking malicious IPs automatically.

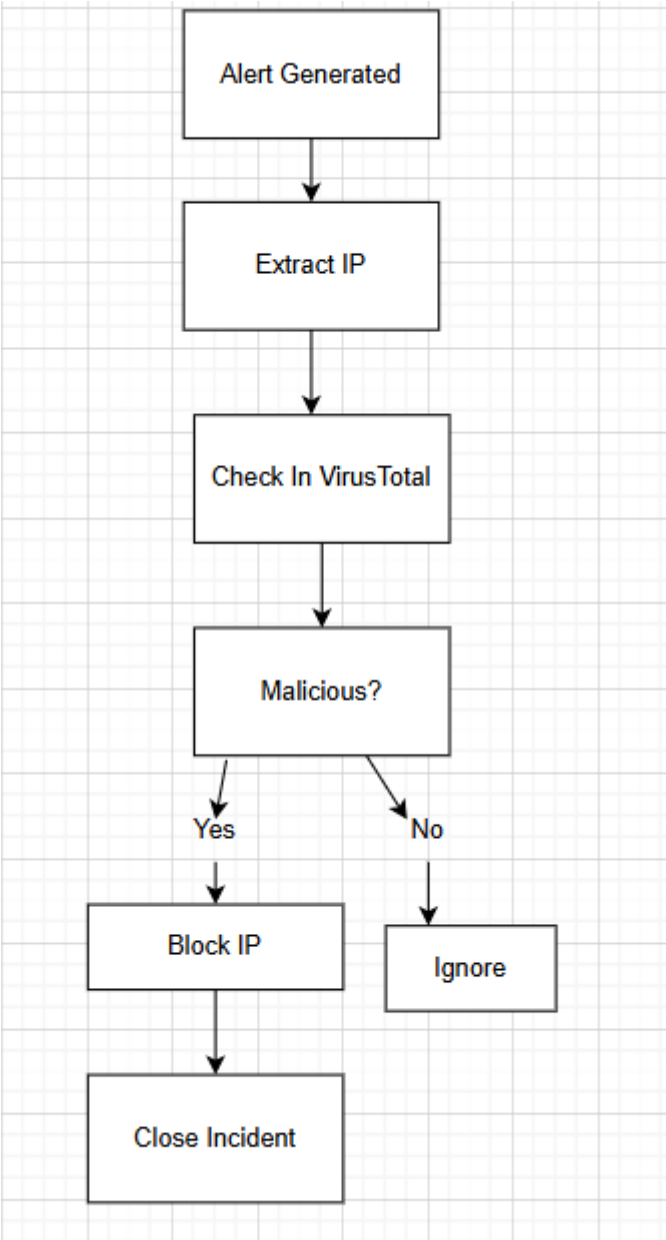


Figure 17: Automated incident response workflow using SOAR

- **Response Actions:-**

  System isolated

  IP blocked

  Case created in TheHive

XIX. Final Report:-

- A complete SOC workflow was implemented by simulating a real-world cyber attack using Metasploit. The attack targeted a vulnerable system using a Samba exploit. The attack was successfully detected using Wazuh, which generated alerts for suspicious activity. The incident was then triaged and analyzed using TheHive platform.
- During the response phase, the affected system was isolated from the network to prevent further damage. The attacker's IP address was blocked using CrowdSec. Automation was applied through a playbook to speed up response actions and reduce manual intervention.
- Post-incident analysis was conducted using root cause analysis techniques, which identified weak system configurations and lack of monitoring rules as key issues. Metrics such as MTTD and MTTR were calculated to evaluate performance.
- Overall, the SOC workflow proved effective in detecting and responding to threats. However, improvements are needed in detection speed, automation, and user awareness training to enhance security capabilities.
- **Conclusion:-**

  This project demonstrates how a SOC handles cyber attacks from detection response. Even in a simulated environment, it highlights the importance of monitoring tools, quick response, and proper security practices.

- **Recommendations:-**

  Regular patch updates.

Use SIEM tools for monitoring.

Implement automated response systems.

Conduct regular security audits.

Improve awareness and training.